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# Circular-Cylinder Wakes with Lattice Boltzmann and Neural Prediction

Week 7 Extension Lecture Notes

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M&I-ENG 690A – AI in Fluid Mechanics

Summer 2026

**Central question.** Can a neural model preserve the convection, phase, strength, and shedding frequency of an unseen cylinder wake, rather than merely produce a visually smooth field?

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# 1 Position in the Course

The original course remains a six-week sequence: Weeks 1–4 establish numerical, kinetic, and surrogate-model foundations; the additive Week 4.1 develops classical POD–Galerkin and POD–DEIM for the cavity; Weeks 5–6 are the controlled research project. This cylinder module is a *Week 7 extension*. It changes the geometry and the dominant physics only after the original sequence is complete.

## Learning checkpoint

By the end of Week 7, a student should be able to:

1. explain the physical transition from attached flow to steady recirculation and periodic vortex shedding;
2. derive the D2Q9 equilibrium distribution and BGK collision update;
3. connect  $Re$ , viscosity, relaxation time, and low-Mach restrictions;
4. measure drag, lift, recirculation length, and Strouhal number;
5. design a complete-case Reynolds holdout without snapshot leakage; and
6. diagnose phase error and downstream vortex diffusion in a learned rollout.

# 2 Cylinder-Wake Physics

For a cylinder of diameter  $D$  in a uniform stream  $U_\infty$ , the governing nondimensional parameter is

$$Re_D = \frac{U_\infty D}{\nu}. \quad (1)$$

At very low  $Re_D$ , the flow is attached and viscous. A symmetric recirculation bubble develops as inertia grows. The steady two-dimensional wake loses stability near  $Re_D \approx 47$ , producing the alternating Bénard–von Kármán vortex street. The precise measured transition and forces depend on blockage, domain extent, discretization, and boundary treatment.

The relevant observables are

$$C_D = \frac{F_D}{\frac{1}{2}\rho U_\infty^2 D}, \quad C_L = \frac{F_L}{\frac{1}{2}\rho U_\infty^2 D}, \quad (2)$$

$$St = \frac{f_s D}{U_\infty}, \quad \omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}. \quad (3)$$

Periodic lift is often the cleanest signal for estimating shedding frequency  $f_s$ . A credible field prediction should reproduce not only pixelwise vorticity but also the spectrum and phase of  $C_L$  and the downstream decay of enstrophy.

# 3 From Boltzmann Kinetics to D2Q9 LBM

The lattice Boltzmann method evolves discrete populations  $f_i(\mathbf{x}, t)$  associated with lattice velocities  $\mathbf{c}_i$ . Density and momentum are moments:

$$\rho = \sum_i f_i, \quad \rho \mathbf{u} = \sum_i f_i \mathbf{c}_i. \quad (4)$$

For D2Q9,

$$\mathbf{c}_i \in \{(0, 0), (\pm 1, 0), (0, \pm 1), (\pm 1, \pm 1)\}, \quad (5)$$

with weights  $w_0 = 4/9$ ,  $w_{1...4} = 1/9$ , and  $w_{5...8} = 1/36$ . The second-order isothermal equilibrium is

$$f_i^{eq} = w_i \rho \left[ 1 + \frac{\mathbf{c}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{c}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u} \cdot \mathbf{u}}{2c_s^2} \right], \quad c_s^2 = \frac{1}{3}. \quad (6)$$

### 3.1 Collision and streaming

The single-relaxation-time BGK update is

$$f_i^*(\mathbf{x}, t) = f_i(\mathbf{x}, t) - \frac{1}{\tau} [f_i(\mathbf{x}, t) - f_i^{eq}(\mathbf{x}, t)], \quad (7)$$

followed by streaming,

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = f_i^*(\mathbf{x}, t). \quad (8)$$

In lattice units,

$$\nu = c_s^2 \left( \tau - \frac{1}{2} \right), \quad \tau = \frac{1}{2} + \frac{U_\infty D}{c_s^2 \text{Re}_D}. \quad (9)$$

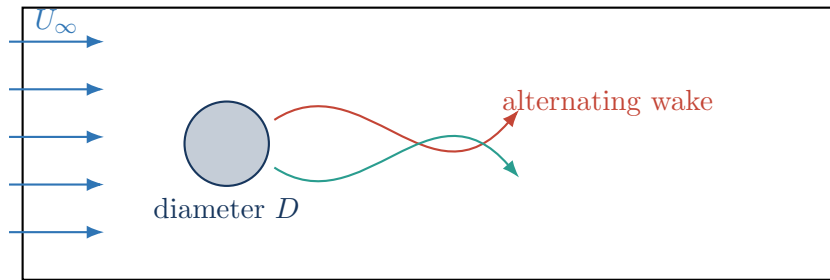
Thus increasing  $\text{Re}_D$  at fixed  $U_\infty$  and  $D$  drives  $\tau$  toward  $1/2$ , where numerical robustness becomes difficult. The notebook provides both transparent BGK and a more robust two-relaxation-time (TRT) option.

#### Important distinction

LBM is weakly compressible. Keep  $\text{Ma} = U_\infty/c_s$  small and monitor density drift. A correct nominal Reynolds number does not compensate for excessive Mach number, inadequate resolution, or boundary contamination.

## 4 Geometry and Boundary Conditions

The computational domain must provide sufficient upstream distance, downstream wake length, and transverse clearance. The cylinder uses a circular solid mask with no-slip bounce-back. The inlet prescribes velocity; the outlet must minimize reflected disturbances; upper and lower boundaries must be consistent with the intended blockage model.



Halfway bounce-back is second-order accurate only for appropriately located, grid-aligned walls. A staircase representation of a circle introduces geometry error; interpolated bounce-back can reduce it. This teaching solver reports the mask and grid resolution explicitly rather than hiding this limitation.

## 5 Force and Frequency Extraction

The momentum-exchange method estimates the force transferred during population reflection. After discarding the transient, compute time-averaged drag, RMS lift, and the dominant nonzero lift-spectrum frequency. Let  $t^* = tU_\infty/D$ ; then the frequency in  $t^*$  is directly  $St$ .

For a shedding case, evidence should include:

1. a sustained  $C_L(t^*)$  oscillation after transient removal;
2. a narrow spectral peak and a stable  $St$  estimate across adjacent windows;
3. alternating signed vorticity in the wake;
4. mean  $C_D$  and  $St$  compared with appropriate two-dimensional references; and
5. sensitivity to grid, domain size, time step, and sampling window.

### Evidence gate

The quick retained cases ( $Re = 5, 20, 40, 100, 180$ ) are teaching evidence. They demonstrate expected regimes and diagnostics, but the qualification path—not a single attractive contour—is the basis for a quantitative claim.

## 6 Learning an Unsteady Wake

An unsteady wake is a transport problem. Vortices translate, deform, and decay. A low-rank linear combination of global modes can smear compact structures when phase is imperfect. Pixelwise mean-square error also rewards smooth predictions, so a network can reduce loss while destroying enstrophy downstream.

### 6.1 Leakage-free split

Snapshots from one trajectory are strongly correlated. Randomly distributing them among training and test sets is leakage. The Week-7 protocol holds out an entire Reynolds-number case. Model selection uses development cases only; the unseen case is opened once after architecture and hyperparameters are frozen.

### 6.2 Four-frame multi-scale CNN

Let four consecutive input fields be

$$\mathcal{X}_n = [\omega_{n-3}, \omega_{n-2}, \omega_{n-1}, \omega_n]. \quad (10)$$

A convolutional model predicts  $\hat{\omega}_{n+1}$ . Multiple spatial scales help represent both the near-cylinder shear layers and larger downstream vortices. Recursive rollout feeds a prediction back as input, which exposes accumulated phase and amplitude error.

Useful evaluation quantities include

$$\epsilon_{L_2}(t) = \frac{\|\hat{\omega}(t) - \omega(t)\|_2}{\|\omega(t)\|_2}, \quad (11)$$

$$E_\omega(x_1, x_2) = \frac{1}{2} \int_{x_1}^{x_2} \int \omega^2 dy dx, \quad (12)$$

$$\epsilon_\phi(t) = \text{wrap} \left( \hat{\phi}(t) - \phi(t) \right). \quad (13)$$

Report the downstream enstrophy ratio, spectral peak, and phase error together with global  $L_2$  error.

## 7 Connection to the 2019 JFM Study

### Connection to Lee & You (JFM, 2019)

Lee and You (2019) developed a deep-learning framework for unsteady circular-cylinder flow prediction using consecutive flow fields, convolutional processing, and recursive future prediction. Their study motivates three principles used here: temporal context must be explicit, evaluation must examine rollout rather than isolated frames, and physical diagnostics are necessary in addition to image similarity.

The present Week-7 exercise is *inspired by*, not a reproduction of, Lee and You. The domain, solver, training set, network capacity, losses, and retained resolution are deliberately smaller for transparent CPU-accessible teaching. Consequently, agreement with one animation cannot be presented as replication of the 2019 result.

The most important comparison is methodological:

| Question            | Lee & You (2019)                         | FlowMLLab Week 7                                                 |
|---------------------|------------------------------------------|------------------------------------------------------------------|
| Temporal input      | Consecutive flow information             | Four consecutive vorticity frames                                |
| Prediction test     | Future-field and recursive behavior      | One-step plus retained blind animation                           |
| Generalization      | Study-specific trained/tested conditions | Complete unseen-Reynolds holdout                                 |
| Physical assessment | Flow evolution beyond visual similarity  | $L_2$ , spectra, phase, divergence/no-slip, downstream enstrophy |
| Purpose             | Research framework                       | Auditable educational extension                                  |

### Important distinction

A sharp first vortex followed by a diffuse downstream wake usually indicates accumulated phase/-transport error, spectral damping, or an MSE-driven smoothing bias. It is not acceptable to judge the model only near the cylinder.

## 8 Notebook Workflow and Decision Gates

The companion notebook follows this order:

1. derive D2Q9 equilibrium and verify moment recovery;
2. audit  $Re$ ,  $Ma$ ,  $\tau$ , blockage, and boundary masks;
3. reproduce the five retained regimes;
4. examine  $C_D$ ,  $C_L$ ,  $St$ , recirculation, and vorticity;
5. establish a POD/phase baseline and observe diffusion failure;
6. freeze the four-frame multi-scale CNN using development cases;
7. open the complete unseen-Reynolds case; and
8. report field error, spectrum, phase, divergence/no-slip, and downstream enstrophy.

**Evidence gate**

Pass only if the model improves on the declared baseline without unacceptable downstream enstrophy loss or spectral/phase failure. If it fails, document the failure; do not retune on the blind case.

**9 Questions for Discussion**

1. Why can a lower global MSE coexist with a visibly worse vortex street?
2. Which error grows first in rollout: amplitude, phase, wavelength, or wake width?
3. How would a vorticity-gradient, spectral, or transport-aware loss change the bias?
4. Why is an unseen Reynolds case stronger evidence than randomly held-out snapshots?
5. Which LBM error could be learned faithfully by the CNN yet remain physically wrong?

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